

ALEX DUNCAN

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PROFESSIONAL EXPERIENCE

LOOKING GLASS FACTORY

Brooklyn, NY | Oct 2016 - Present

Leadership across software product, engineering, creator tools, and hands-on development for glasses-free holographic displays.

VP of Software Product

Nov 2024 - Present

- Lead software product strategy across consumer applications, embodied AI, developer platforms, and enterprise workflows, balancing market, customer, hardware, and engineering constraints from discovery through launch.
- Direct product and UX for musubi; adapted computer-vision models for private, cross-platform ONNX inference and shaped the workflow that converts photos and videos into media for a new non-light-field holographic display category.
- Lead Liteforms, an embodied conversational-AI platform combining LLMs, RAG, speech, animation, and spatial rendering, alongside the company's internal AI adoption and governance work.
- Work across engineering, design, sales, customer success, and executive leadership; engage directly with organizations including NASA, NVIDIA, Intel, Accenture, and the US Space Force to translate institutional needs into roadmaps, pilots, proposals, and launch criteria.

Team Lead, Creator Tools

Mar 2021 - Nov 2024

- Owned the creator and developer ecosystem across Unity, Unreal Engine, Blender, WebXR, native SDKs, documentation, samples, workshops, onboarding, and technical support.
- Led discovery, user research, product design, launch, and iteration for creator tools and applied-AI products; developed partner scopes and trained external technical and creative teams.
- Managed and mentored engineers and designers working across ambiguous hardware, software, customer, and go-to-market problems.

VP of Software Engineering

Jun 2019 - Mar 2021

- Led the software engineering organization and full software suite required to launch Looking Glass Portrait and the Looking Glass 32-inch display, including architecture, hiring, planning, and release management.
- Helped scale the company to more than 50 people while turning experimental light-field technology into repeatable products and customer workflows.

Engineering Manager / Software Developer

Oct 2016 - Jun 2019

- Introduced the software development lifecycle, Agile planning, engineering standards, and structured architectural review while continuing hands-on product development.
- Built spatial applications, renderers, web and 3D-model tools in Unity/C#, Python, GLSL, JavaScript, and three.js; invented a depth-aware 3D cursor interaction reflected in the patent portfolio.

Adjunct Professor | Long Island University

Jan 2017 - May 2017

- Designed and taught Games Through History, connecting game mechanics and technologies to their cultural, political, and economic contexts.

SELECTED PATENTS

Augmented Superstereoscopic Display

2026

US Patent 12,587,629. Co-inventor. Combines a multiview 3D display with an additional image layer so crisp text and interface elements can coexist with spatial content.

System and Method for Generating and Interacting with Conversational Three-Dimensional Subjects

2024

US Application 18/610,787, pending; published as US 2024/0323332 A1. Co-inventor. Covers embodied conversational systems coordinating language models, history, sensors, voice, animation, and character behavior.

System and Method for Displaying a Three-Dimensional Image

2023

US Patent 11,736,680. Co-inventor. Describes a reliable pipeline from 2D, depth, or 3D source material through multiview rendering and verified device calibration to display-ready light-field imagery.

EDUCATION & PROFESSIONAL DEVELOPMENT

New York University - MFA, Game Design

2016

Graduate study in game design, interaction design, creative technology, critical theory, prototyping, and play research.

University of British Columbia - BA, English Literature (Honours); BA, Economics

2013

Interdisciplinary training in textual analysis, research and argument, political economy, institutions, and quantitative reasoning.

Additional study: AI Safety Fundamentals: AI Governance, BlueDot Impact | Justice and Equity in Technology Policy, University of Michigan | Data Analysis with Python, IBM | Introduction to SQL, University of Michigan

SERVICE & TEACHING

POWRPLNT - Governing Board Member, Secretary & Fundraising Committee Lead

2020

Provide governance and fundraising support to a Brooklyn nonprofit expanding access to digital art and technology for underrepresented young adults.

Collège François Mauriac - Volunteer Game Design Instructor

2015

Taught middle-school students in Bordeaux to design and build 2D platform games using their own visual assets.

SELECTED PRODUCT WORK

musubi

2026

Product lead for a cross-platform desktop app that privately converts photos and short videos into display-ready holographic media. Adapted AI models for local ONNX inference and designed editing, album, preview, and device-transfer workflows for a new Hololuminescent Display category.

Parsem: AI Search Redesigned

2025

Creator. AI search interface that maps each part of an answer to a visible source passage, preserves provenance, and directs attention back to original reporting.

Surely, AI Can

2025

Creator. Public-facing AI benchmarking and analysis project focused on surprising frontier-model failures and the gap between demonstrations and reliable deployment.

Liteforms

2023

Product lead for a desktop application for talking to and creating holographic AI characters, combining LLMs, RAG, voice, animation, and shared spatial presence.

Looking Glass Unity Plugin

2019

Product lead for a creator tool that translated light-field rendering into familiar Unity concepts: a volumetric camera, quilt capture, depth-aware focus, and spatial interaction.

On Nature

2017

Creator. Web game that renders one environment through several representational modes, examining how visual systems construct ideas of nature.

Standard Assets

2017

Creator. Experimental 3D game examining how Unity's default environment package encodes conventions about landscape, agency, and play.

Every Creeping Thing

2016

Creator. Three-part atmospheric game using unusual rendering and control systems to explore animal perception and the limits of empathetic simulation.

DESIGNxBOULDER

2016

Developer. Interactive environmental visualization for artist Natalie Jeremijenko and the Boulder Museum of Contemporary Art.

Investing in Futures

2016

Game-design contributor. Card game by More&More Unlimited that prompts players to imagine alternative global futures.

SELECTED WRITING & PUBLICATIONS

- 'Beyond Fair Use: Sustaining Journalism in the Age of AI.' 2025. Research essay on AI's dependence on journalism, the economics of news production, copyright and bargaining frameworks, and policy options for sustaining public-interest information.
- 'Savage Beasts: The Spatial Conflict Between Civilization and Wilderness in The Elder Scrolls V: Skyrim.' First Person Scholar, Feb 14, 2018. Publication
- 'Finding Wriggle Room: How I Made a Maggot Simulator.' A MAZE. Magazine, Issue 4, Aug 2016. Publication
- 'When Lions Speak: Representations of Non-Human Avatars in Digital Games.' Paper examining animal subjectivity and interface design in Shelter and Spore.
- 'Games Biting Back: ANATOMY and Ecofeminism.' Published abstract, Abstract Proceedings of DiGRA 2018 Conference: The Game is the Message. DOI
- 'Time is Money: Cart Life and the Instrumentalization of Time.' Paper on how game systems critique the economic instrumentalization of everyday time.
- 'Ladies' Man: Womanizing in The Witcher.' The Ontological Geek. Archived publication

SELECTED CONFERENCE PRESENTATIONS

- 'The Looking Glass: A Holographic Platform for 3D Creators.' FITC Toronto 2019, Toronto, Canada, Apr 30, 2019. Forty-five-minute presentation on glasses-free, shared 3D; the Looking Glass platform; and creator workflows spanning Unity, Three.js, Blender, and Maya.
- 'Games Biting Back: ANATOMY and Ecofeminism.' DiGRA 2018: The Game is the Message, University of Turin, Turin, Italy, Jul 25-28, 2018.
- 'Time is Money: Cart Life and the Instrumentalization of Time.' Extending Play, Rutgers University, Sep 2016.
- 'When Lions Speak: Representations of Non-Human Avatars in Digital Games.' DiGRA/FDG 2016, Abertay University, Dundee, Scotland, Aug 1-6, 2016.
- 'When Lions Speak: Representations of Non-Human Avatars in Digital Games.' Games and Literary Theory, Loyola University New Orleans, Nov 20-22, 2015.